

Solution to exercises in Lecture 1

Q1(a) Navier - Stokes equation :

$$\rho \frac{\partial u}{\partial t} + \underbrace{\rho (u \cdot \nabla) u}_{\text{non-linear due to this term}} = -\nabla p + \nabla \cdot (\eta [(\nabla u) + (\nabla u)^T])$$

non-linear due to this term

(b) Cahn - Hilliard equation :

$$\frac{\partial c}{\partial t} = D \underbrace{\nabla^2 (c^3 - c - \gamma \nabla^2 c)}_{\text{non-linear due to this term}}$$

non-linear due to this term

(c) Heat equation

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u$$

linear as both terms have 1st power in derivatives of u

Q2) (i) rewrite as : $-\kappa \frac{\partial^2 T}{\partial x^2} - \kappa \frac{\partial^2 T}{\partial y^2} + \frac{\partial T}{\partial t} = 0$

map to : $a_{tt} T_{tt} + a_{xx} T_{xx} + a_{yy} T_{yy} + \dots = 0$

Note that (i) we have no $\frac{\partial^2 T}{\partial t^2}$ term so $a_{tt} = 0$

(ii) we have no $\frac{\partial^2 T}{\partial t \partial x}$, $\frac{\partial^2 T}{\partial x \partial y}$, $\frac{\partial^2 T}{\partial t \partial y}$ terms either

Hence $[a_{ij}] = \begin{pmatrix} a_{tt} & a_{tx} & a_{ty} \\ a_{xt} & a_{xx} & a_{xy} \\ a_{yt} & a_{yx} & a_{yy} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\kappa & 0 \\ 0 & 0 & -\kappa \end{pmatrix} \Rightarrow \text{parabolic PDE}$
 $\det [a_{ij}] = 0$

Alternatively, we can check the number of positive, negative & zero eigenvalues.
For parabolic PDE $\det [a_{ij}] \neq 0$ means at least one of the eigenvalues must be zero.

Q2) ii) Rewrite as : $\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$

map to : $\underbrace{a_{tt}}_{=1} u_{tt} + \underbrace{a_{xx}}_{=-c^2} u_{xx} + \underbrace{a_{tx}}_{=0} u_{tx} + \underbrace{a_{xt}}_{=0} u_{xt} = 0$

so $[a_{ij}] = \begin{pmatrix} 1 & 0 \\ 0 & -c^2 \end{pmatrix} \Rightarrow$ hyperbolic PDE
 $\underbrace{z=0}_{\text{no of zero eigenvalue}}, \underbrace{p=1}_{\text{eigenvalue}}, \underbrace{N=1}_{\text{eigenvalue}}$

Alternatively, we can calculate $a_{tx}^2 - a_{tt} a_{xx} = 0^2 - (1)(-c^2)$

$a_{tx}^2 - a_{tt} a_{xx} = c^2 > 0$
 hyperbolic PDE

Q3) $u_{xx} = \frac{u(x+\Delta x) + u(x-\Delta x) - 2u(x)}{\Delta x^2}$

$= \frac{1}{\Delta x^2} \left[\underbrace{u + u - 2u}_{\text{cancels}} + \underbrace{\Delta x \frac{\partial u}{\partial x} - \Delta x \frac{\partial u}{\partial x}}_{\text{cancels}} + \frac{\Delta x^2}{2!} \frac{\partial^2 u}{\partial x^2} + \underbrace{\frac{\Delta x^3}{3!} \frac{\partial^3 u}{\partial x^3} - \frac{\Delta x^3}{3!} \frac{\partial^3 u}{\partial x^3}}_{\text{cancels}} + \frac{\Delta x^4}{4!} \frac{\partial^4 u}{\partial x^4} + \dots \right]$

$= \underbrace{\frac{\partial^2 u}{\partial x^2}}_{\text{what we want}} + \underbrace{\frac{2\Delta x^4}{4!} \frac{\partial^4 u}{\partial x^4} + \dots}_{\text{leading error term}}$

what we want

leading error term
 2nd order because $\frac{\partial^4 u}{\partial x^4}$ is 2 orders higher in derivatives compared to $\frac{\partial^2 u}{\partial x^2}$

Q4) The maths to derive u_x & u_{xx} are the same because they don't involve any expansion in Δt . So 1D & 2D derivations must give the same result.

The derivations for u_t & u_{tx} are the same for u_x & u_{xx} except we swap the x & t labels.